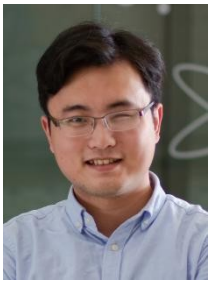
Shi-Ju Ran Email: sjran@cnu.edu.cn



Professional Profile: Professor and doctoral supervisor in the Department of Physics at Capital Normal University; currently Deputy Director for Research. Research interests include quantum artificial intelligence, quantum many-body physics, tensor-network theory and methods, and quantum information and quantum computing. Teaches the undergraduate major courses Electrodynamics and Methods of Mathematical Physics, and the required course Introduction to Artificial Intelligence A. Has published more than 60 papers, including articles in Physical Review Letters; work on tree-structured unitary tensor-network machine learning received an international highly cited paper award. First author of the English monograph Tensor Network Contractions (Springer), which received a positive review in a leading international mathematics journal; sole author of the Chinese monograph Tensor Networks, supported by the China National Publication Fund. Inventor on four granted Chinese invention patents. Principal Investigator of an NSFC Young Scientists Fund project, Beijing Natural Science Foundation General Program projects, a Beijing Municipal Education Commission project, and a CAS program subproject; key participant in an NSFC Key Program. Referee for Physical Review Letters and Nature portfolio journals, and an invited speaker at numerous international conferences. Former guest editor for Q1/Q2 SCI journals, committee member for international machine-learning conferences, and project reviewer for a European academic committee. Selected for a Beijing overseas talent program and received a Beijing Outstanding Supervisor Award. Executive Committee Member of the CCF Technical Committee on Quantum Computing.

Student Recruitment: AI + Physics. Students interested in interdisciplinary research at the interface of AI and physics are welcome to join the group. Current directions include large AI models for theoretical physics (machine-assisted derivation, numerical computation, and related topics), integration with quantum computing, and tensor-network machine-learning methods and their interpretability.

Education, Research, and Employment:

Sep 2006 - Jul 2010, B.Sc. in Physics, Beijing Normal University;

Sep 2010 - Jul 2015, Ph.D. in Physics, University of Chinese Academy of Sciences (Advisor: Prof. Gang Su);

Jul 2014, Visiting Ph.D. Student, Ludwig Maximilian University of Munich, Germany;

Jul 2015 - Jul 2018, Postdoctoral Researcher, ICFO - The Institute of Photonic Sciences, Spain (Host: Prof. Maciej Lewenstein);

2016, Fundacio Catalunya Independent Postdoctoral Fellowship;

Sep 2017, Visiting Scholar, Max Planck Institute of Quantum Optics, Germany;

Jun 2018, Visiting Scholar, Johannes Gutenberg University Mainz, Germany;

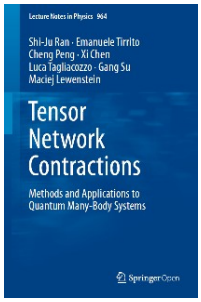
Jul - Nov 2018, Visiting Scholar, University of Chinese Academy of Sciences;

Nov - Dec 2018, Lecturer, Department of Physics, Capital Normal University;

Jan 2019 - Dec 2023, Associate Professor, Department of Physics, Capital Normal University;

Jan 2024 - Present, Professor, Department of Physics, Capital Normal University;

Academic Monographs:



- **Shi-Ju Ran***, Emanuele Tirrito, Cheng Peng, Xi Chen, Luca Tagliacozzo, Gang Su, and Maciej Lewenstein, "Tensor Network Contractions", *Lecture Notes in Physics, Springer, Cham* (2020). ISSN: 0075-8450; ISBN: 978-3-030-34488-7. DOI: https://doi.org/10.1007/978-3-030-34489-4

This book provides a systematic introduction to tensor-network methods in quantum physics and summarizes a series of the authors' original contributions. According to official Springer data, it has received more than 100,000 downloads worldwide.



- **Shi-Ju Ran**, Tensor Networks, Capital Normal University Press, Beijing (2022). ISBN: 978-7-5656-7150-0

The first Chinese-language monograph on tensor networks, it develops the subject from basic tensor definitions through quantum physics, statistics, tensor foundations, and recent methods in quantum physics and machine learning. It is intended as a systematic introduction for physics undergraduates, graduate students, researchers, and readers in related fields. Supported by the 2023 China National Publication Fund.

Academic Publications:

[1] Da-zhi Fang, **Shi-Ju Ran***, and Gang Su*, Exotic transition and reentrance on the 1/9 magnetization plateau in a spin-1/2 anisotropic kagome antiferromagnet *Phys. Rev. B* **113**, L140405 (10 April 2026).

[2] Shengchen Bai, Kun Zhang, and **Shi-Ju Ran***, Orthogonality in Quantum-Probabilistic Machine Learning: An Investigation on Multi-qubit Encoding, Intell Comput. 5, 0232 (24 March 2026).

[3] Hao Zhu, Ding-Zu Wang*, **Shi-Ju Ran***, and Guo-Feng Zhang*, Statistics-encoded tensor network approach in disordered quantum many-body spin chains, *Phys. Rev. B* **113**, 085123 (11 February 2026).

[4] Sheng-Chen Bai and **Shi-Ju Ran***, Machine learning of chaotic characteristics in classical nonlinear dynamics using variational quantum circuit, *Chinese Phys. B* **35** 020303 (2026).

[5] Shuo Qi, Wen-Jun Li*, Gang Su*, and **Shi-Ju Ran***, Matrix-Product Entanglement Characterizing the Optimality of State-Preparation Quantum Circuits, *Phys. Rev. Lett.* **136**, 020602 (14 January 2026).

[6] Wenbo Qiao, Peng Zhang*, Jiaming Zhao, and **Shi-Ju Ran**, Unveiling the nature of graphs through quantum graphon learning, *npj Quantum Inf.*, **12**, 8 (8 January 2026).

[7] Rui Hong, Hao-Wei Cui, An-Chun Ji, and **Shi-Ju Ran***, Entanglement scaling and criticality of infinite-size quantum many-body systems in continuous space addressed by a tensor network approach, *Phys. Rev. B* **111**, 245109 (3 June 2025).

[8] Yong Qing, Ke Li, Peng-Fei Zhou, and **Shi-Ju Ran***, Compressing Neural Networks Using Tensor Networks with Exponentially Fewer Variational Parameters, *Intell. Comput.* **4**, 0123

(15 May 2025).

[9] Sheng-Chen Bai and **Shi-Ju Ran***, Universal Scaling Laws in Quantum-Probabilistic Machine Learning by Tensor Network: Toward Interpreting Representation and Generalization Powers, *Chin. Phys. Lett.* **41**, 120701 (19 November 2024).

[10] Ying Lu, Pei Shi, Xiao-Han Wang, Jie Hu, and **Shi-Ju Ran***, Persistent Ballistic Entanglement Spreading with Optimal Control in Quantum Spin Chains, *Phys. Rev. Lett.* **133**, 070402 (13 August 2024).

[11] Ding-Zu Wang, Hao Zhu, Jian Cui, Javier Argüello-Luengo, Maciej Lewenstein, Guo-Feng Zhang*, Piotr Sierant*, and **Shi-Ju Ran***, Eigenstate thermalization and its breakdown in quantum spin chains with inhomogeneous interactions, *Phys. Rev. B* **109**, 045139 (22 January 2024).

[12] Ding-Zu Wang, Guo-Feng Zhang*, Maciej Lewenstein*, and **Shi-Ju Ran***, Boundary-induced singularity in strongly-correlated quantum systems at finite temperature, *Quantum Science and Technology* **9**, 015008 (2024).

[13] **Shi-Ju Ran*** and Gang Su*, Tensor network for interpretable and efficient quantum-inspired machine learning, *Intelligent Computing* **2**, 0061 (17 Nov 2023).

[14] Ying Lu and **Shi-Ju Ran***, Many-body control with reinforcement learning and tensor networks, *Nature Machine Intelligence*, News&Views (4 October 2023).

[15] Peng-Fei Zhou, Ying Lu, Jia-Hao Wang, and **Shi-Ju Ran***, Tensor Network Efficiently Representing Schmidt Decomposition of Quantum Many-Body States, *Phys. Rev. Lett.* **131**, 020403 (13 July 2023).

[16] Da-Zhi Fang, Ning Xi, **Shi-Ju Ran***, and Gang Su*, Nature of the 1/9-magnetization plateau in the spin-1/2 kagome Heisenberg antiferromagnet, *Phys. Rev. B* **107**, L220401 (13 June 2023).

[17] Ye-Ming Meng, Jing Zhang, Peng Zhang, Chao Gao*, and **Shi-Ju Ran***, Residual matrix product state for machine learning, *SciPost Phys.* **14**, 142 (02 June 2023).

[18] Ying Lu, Peng-Fei Zhou, Shao-Ming Fei*, and **Shi-Ju Ran***, Quantum compiling with a variational instruction set for accurate and fast quantum computing, *Phys. Rev. Research* **5**, 023096 (12 May 2023).

[19] Shi-Ju Ran and Gang Su, Tensor-Network Quantum Machine Learning: Interpretations of Space, Measure, and Probability, Communications of the CCF 19(4), 17-21 (2023).

[20] Peng Zhang and Shi-Ju Ran (Guest Editors), Quantum Artificial Intelligence, Communications of the CCF 19(4), 8-9 (2023).

[21] **Shi-Ju Ran***, Qibin Zhao, Peng Zhang, and Chu Guo, Editorial: Tensor network approaches for quantum many-body physics and machine learning, *Frontiers in Physics* **11**, 1170492 (27 February 2023).

[22] Sheng-Chen Bai, Yi-Cheng Tang, and **Shi-Ju Ran***, Unsupervised Recognition of Informative Features via Tensor Network Machine Learning and Quantum Entanglement Variations, *Chinese Phys. Lett.* **39**, 100701 (15 September 2022).

[23] Ding-Zu Wang, Guo-Feng Zhang*, Maciej Lewenstein*, and **Shi-Ju Ran***, Efficient simulation of quantum many-body thermodynamics by tailoring a zero-temperature tensor network, *Phys. Rev. B* **105**, 155155 (27 April 2022).

[24] Rui Hong, Ya-Xuan Xiao, Jie Hu, An-Chun Ji, and **Shi-Ju Ran***, Functional tensor network solving many-body Schrödinger equation, *Phys. Rev. B* **105**, 165116 (12 April 2022).

[25] Wei-Ming Li and **Shi-Ju Ran***, Non-Parametric Semi-Supervised Learning in Many-Body Hilbert Space with Rescaled Logarithmic Fidelity, *Mathematics* **10**, 940 (15 March 2022).

[26] Monika Aidelsburger, Luca Barbiero, Alejandro Bermudez, Titas Chanda, Alexandre Dauphin, Daniel González-Cuadra, Przemysław R. Grzybowski, Simon Hands, Fred Jendrzejewski, Johannes Jünemann, Gediminas Juzeliunas, Valentin Kasper, Angelo Piga, **Shi-Ju Ran**, Matteo Rizzi, Germán Sierra, Luca Tagliacozzo, Emanuele Tirrito, Torsten V. Zache, Jakub Zakrzewski, Erez Zohar, and Maciej Lewenstein*, Cold atoms meet lattice gauge theory, *Phil. Trans. R. Soc. A* **380**, 20210064 (20 December 2021).

[27] Ying Lu, Yue-Min Li, Peng-Fei Zhou, and **Shi-Ju Ran***, Preparation of Many-body Ground States by Time Evolution with Variational Microscopic Magnetic Fields and Incomplete Interactions, *Phys. Rev. A* **104**, 052413 (11 November 2021).

[28] Kunkun Wang, Lei Xiao, Wei Yi*, **Shi-Ju Ran***, and Peng Xue*, Experimental realization of a quantum image classifier via tensor-network-based machine learning, *Photon. Res.* **9**, 2332-2340 (8 November 2021) (Editor's pick).

[29] Peng-Fei Zhou, Rui Hong, and **Shi-Ju Ran***, Automatically differentiable quantum circuit for many-qubit state preparation, *Phys. Rev. A* **104**, 042601 (1 October 2021).

[30] Xinran Ma, Z. C. Tu, and **Shi-Ju Ran***, Deep Learning Quantum States for Hamiltonian Estimation, *Chin. Phys. Lett. (Express Letter)* **38** (11), 110301 (11 October 2021) (Cover Article).

[31] Rui Hong, Peng-Fei Zhou, Bin Xi, Jie Hu, An-Chun Ji, and **Shi-Ju Ran***, Predicting quantum potentials by deep neural network and metropolis sampling, *SciPost Physics Core* **4**, 022 (13 September 2021).

[32] Yuhan Liu, Wen-Jun Li, Xiao Zhang, Maciej Lewenstein, Gang Su*, and **Shi-Ju Ran***, Entanglement-Based Feature Extraction by Tensor Network Machine Learning, *Front. Appl. Math. Stat.* **7**, 716044 (06 August 2021).

[33] Yuan Yang, Zheng-Zhi Sun, **Shi-Ju Ran***, and Gang Su*, Visualizing quantum phases and identifying quantum phase transitions by nonlinear dimensional reduction, *Phys. Rev. B* **103**, 075106 (2 February 2021).

[34] Yuan Yang, Zhengchuan Wang*, **Shi-Ju Ran***, and Gang Su*, Phase identification in many-body systems by virtual configuration binarization, *Phys. Rev. E* **103**, 013313 (22 January 2021).

[35] **Shi-Ju Ran***, Zheng-Zhi Sun, Shao-Ming Fei, Gang Su, and Maciej Lewenstein, Tensor network compressed sensing with unsupervised machine learning, *Phys. Rev. Research* **2**, 033293 (24 August 2020).

[36] Zheng-Zhi Sun, **Shi-Ju Ran*,** and Gang Su*, Tangent-Space Gradient Optimization of Tensor Network for Machine Learning, *Phys. Rev. E* **102**, 012152 (30 July 2020).

[37] **Shi-Ju Ran**, Encoding of matrix product states into quantum circuits of one- and two-qubit gates, *Phys. Rev. A* **101**, 032310 (9 March 2020).

[38] Zheng-Zhi Sun, Cheng Peng, Ding Liu, **Shi-Ju Ran***, and Gang Su*, Generative Tensor Network Classification Model for Supervised Machine Learning, *Phys. Rev. B* **101**, 075135 (25 February 2020).

[39] Yuan Yang, **Shi-Ju Ran***, Xi Chen, Zheng-zhi Sun, Shou-Shu Gong, Zhengchuan Wang*, and Gang Su*, Reentrance of Topological Phase in Spin-1 Frustrated Heisenberg

Chain, *Phys. Rev. B* **101**, 045133, (29 January 2020).

[40] **Shi-Ju Ran***, Bin Xi, Cheng Peng, Gang Su, and Maciej Lewenstein, Efficient quantum simulation for thermodynamics of infinite-size many-body systems in arbitrary dimensions, *Phys. Rev. B* **99**, 205132 (20 May 2019).

[41] Ding Liu, **Shi-Ju Ran***, Peter Wittek*, Cheng Peng, Rual Blázquez Garca, Gang Su, and Maciej Lewenstein, Machine Learning by Unitary Tensor Network of Hierarchical Tree Structure, *New J. Phys.* **21**, 073059 (30 July 2019).

[42] Xi Chen, **Shi-Ju Ran***, Shuo Yang, Maciej Lewenstein, Gang Su*, Noise-tolerant Signature of Z_N Topological Orders in Quantum Many-body States, *Phys. Rev. B* **99**, 195101 (1 May 2019).

[43] Cheng Peng and Shi-Ju Ran, Time Matrix Product State: Theory and Applications, Journal of University of Chinese Academy of Sciences 36(6) (2019).

[44] **Shi-Ju Ran***, Cheng Peng, Gang Su, and Maciej Lewenstein, Controlling the phase diagram of finite spin-1/2 chains by tuning the boundary interactions, *Phys. Rev. B* **98**, 085111 (7 August 2018).

[45] **Shi-Ju Ran**, Wei Li, Shou-Shu Gong, Andreas Weichselbaum, Jan von Delft, and Gang Su*, Emergent spin-1 trimerized valence bond crystal in the spin-1/2 Heisenberg model on the star lattice, *Phys. Rev. B* **97**, 075146 (26 February 2018).

[46] Cheng Peng, **Shi-Ju Ran***, Maciej Lewenstein, and Gang Su*, Exotic entanglement scaling of Heisenberg antiferromagnet on honeycomb lattice, *Eur. Phys. J. B* **91**, 258 (22 October 2018).

[47] Xi Chen, **Shi-Ju Ran**, Tao Liu, Cheng Peng, Yi-Zhen Huang, and Gang Su*, Thermodynamics of spin-1/2 Kagomé Heisenberg antiferromagnet: algebraic paramagnetic liquid and finite-temperature phase diagram, *Sci. Bull.* **63**, 1545–1550 (22 November 2018).

[48] **Shi-Ju Ran***, Angelo Piga, Cheng Peng, Gang Su, and Maciej Lewenstein, Few-body systems capture many-body physics: Tensor network approach, *Phys. Rev. B* **96**, 155120 (13 October 2017).

[49] **Shi-Ju Ran**, Cheng Peng, Wei Li, Maciej Lewenstein, and Gang Su*, Criticality in two-dimensional quantum systems: Tensor network approach, *Phys. Rev. B* **95**, 155114 (10 April 2017).

[50] J. Jünemann, A. Piga, **Shi-Ju Ran**, M. Lewenstein, M. Rizzi, and A. Bermudez*, Exploring Interacting Topological Insulators with Ultracold Atoms: the Synthetic Creutz-Hubbard Model, *Phys. Rev. X* **7**, 031057 (27 September 2017).

[51] Cheng Peng, **Shi-Ju Ran**, Tao Liu, Xi Chen, and Gang Su*, Fermionic algebraic quantum spin liquid in an octa-kagome frustrated antiferromagnet, *Phys. Rev. B* **95**, 075140 (22 February 2017).

[52] Emanuele Tirrito, **Shi-Ju Ran**, Andrew J Ferris, Ian P McCulloch, and Maciej Lewenstein*, Efficient perturbation theory to improve the density matrix renormalization group, *Phys. Rev. B* **95**, 064110 (21 February 2017).

[53] **Shi-Ju Ran**, Ab-initio optimization principle for the ground states of translationally invariant strongly correlated quantum lattice models, *Phys. Rev. E* **93**, 053310 (27 May 2016).

[54] Meng Wang, **Shi-Ju Ran**, Tao Liu, Yang Zhao, Qing-Rong Zheng, and Gang Su*, Phase diagram and exotic spin-spin correlations of anisotropic Ising model on the Sierpiński gasket, *Eur. Phys. J. B* **89**, 1-10 (1 February 2016).

[55] Tao Liu, **Shi-Ju Ran**, Wei Li, Xin Yan, Yang Zhao, and Gang Su*, Featureless quantum spin liquid, 1/3-magnetization plateau state, and exotic thermodynamic properties of the spin-1/2 frustrated Heisenberg antiferromagnet on an infinite Husimi lattice, *Phys. Rev. B* **89**, 054426 (24 February 2014).

[56] **Shi-Ju Ran**, Bin Xi, Tao Liu, and Gang Su*, Theory of network contractor dynamics for exploring thermodynamic properties of two-dimensional quantum lattice models, *Phys. Rev. B* **88**, 064407 (12 August 2013).

[57] Yang Zhao, Wei Li, Bin Xi, Zhe Zhang, Xin Yan, **Shi-Ju Ran**, Tao Liu, and Gang Su*, Kosterlitz-Thouless phase transition and re-entrance in an anisotropic three-state Potts model on the generalized kagome lattice, *Phys. Rev. E* **87**, 032151 (22 March 2013).

[58] Yang Zhao, Wei Li, Bin Xi, **Shi-Ju Ran**, Yuan-Yuan Zhu, Bing-Wu Wang, Song Gao, and Gang Su*, Honeycomb Heisenberg spin ladder: Unusual ground state and thermodynamic properties, *Eur. Phys. Lett.* **104**, 57009 (23 December 2013).

[59] **Shi-Ju Ran**, Wei Li, Bin Xi, Zhe Zhang, and Gang Su*, Optimized decimation of tensor networks with super-orthogonalization for two-dimensional quantum lattice models, *Phys. Rev. B* **86**, 134429 (26 October 2012).

[60] Xin Yan, Wei Li, Yang Zhao, **Shi-Ju Ran**, and Gang Su*, Phase diagrams, distinct conformal anomalies, and thermodynamics of spin-1 bond-alternating Heisenberg antiferromagnetic chain in magnetic fields, *Phys. Rev. B* **85**, 134425 (13 April 2012).

[61] Wei Li, **Shi-Ju Ran**, Shou-Shu Gong, Yang Zhao, Bin Xi, Fei Ye, and Gang Su*, Linearized tensor renormalization group algorithm for the calculation of thermodynamic properties of quantum lattice models, *Phys. Rev. Lett.* **106**, 127202 (22 March 2011).

[62] Wei Li, Shou-Shu Gong, Yang Zhao, **Shi-Ju Ran**, Song Gao, and Gang Su*, Phase transitions and thermodynamics of the two-dimensional Ising model on a distorted kagome lattice, *Phys. Rev. B* **82**, 134434 (21 October 2010).

Granted Invention Patents:

- Chinese invention patent: Neural Network Compression Method, Device, Electronic Equipment, and Storage Medium; Patent No.: ZL 2023 1 0460357.7; Inventors: Shi-Ju Ran, Yong Qing, Ke Li, and Peng-Fei Zhou; Country of grant: People's Republic of China; Grant date: 14 Jul 2023.
- Chinese invention patent: Email Processing Method, Device, Electronic Equipment, and Storage Medium; Patent No.: ZL 2025 1 0765361.3; Inventors: Shi-Ju Ran and Sheng-Chen Bai; Country of grant: People's Republic of China; Grant date: 16 Sep 2025.
- Chinese invention patent: Method for Optimizing Artificial Intelligence Models Based on Tensor Structures; Patent No.: ZL 2026 1 0416463.9; Inventors: Gang Su, Ying Lu, Shi-Ju Ran, and Peng-Fei Zhou; Country of grant: People's Republic of China; Grant date: 30 Jun 2026.
- Chinese invention patent: Electrocardiogram Recognition Method, Device, Electronic Equipment, and Storage Medium; Patent No.: ZL 2026 1 0087605.1; Inventors: Shi-Ju Ran, Kun Zhang, Sheng-Chen Bai, Lei Ding, and Min Tang; Country of grant: People's Republic of China; Grant date: 8 Sep 2026.

Conferences Organized:

- 23-28 Aug 2026, Third Workshop on Quantum Intelligent Computing and First "Guangliang Qingcheng" Summer School on Quantum Intelligent Computing, Dujiangyan, Sichuan; organizers: Key Laboratory of Quantum Physics and Photonic Quantum Information, Ministry of Education; Capital Normal University; and Tianjin

University; organizing committee contacts: Zizhu Wang, Shi-Ju Ran, and Peng Zhang.

- 10-12 Jan 2025, Second Workshop on Quantum Intelligent Computing, Dujiangyan, Sichuan; organizers: University of Electronic Science and Technology of China, Capital Normal University, and Tianjin University; organizing committee: Zizhu Wang, Shi-Ju Ran, and Peng Zhang.
- 21-22 Nov 2020, Workshop on Quantum Physics and Intelligent Computing, online; organizers: Tianjin University and Capital Normal University; organizing committee: Peng Zhang and Shi-Ju Ran.

Invited Talks and Conference Papers (2020-Present):

[1] Invited talk (Chinese): **Shi-Ju Ran**, Quantum-Inspired Optimization Methods for Quantum Many-Body Problems, 2026 Winter School on Artificial Intelligence and Quantum Technology, Hetao Institute, Shenzhen (22 Jan 2026).

[2] Invited talk (Chinese): **Shi-Ju Ran**, Quantum-Inspired Machine Learning Based on Tensor Networks, "LLM and Science" Program, Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing (12 Dec 2024).

[3] Invited talk (English): **Shi-Ju Ran**, Quantum-Inspired Machine Learning Based on Tensor Networks, 1st Quantum Intelligence Workshop, Institute of Physics, Chinese Academy of Sciences, Beijing (1 Dec 2025).

[4] Invited talk (Chinese): **Shi-Ju Ran**, New Tensor-Network Methods for Quantum Many-Body Physics at the Intersection of Machine Learning and Quantum Computing, 204th Science and Technology Frontier Forum: Scientific Foundations of Artificial Intelligence, Academic Divisions of the Chinese Academy of Sciences, Beijing (30 Nov 2025).

[5] Invited talk (Chinese): **Shi-Ju Ran**, Quantum Many-Body Approaches Based on Variational Quantum Circuits and Tensor Networks, 22nd National Conference on Condensed Matter Theory and Statistical Physics, South China Normal University (8 Nov 2025).

[6] Invited talk (Chinese): **Shi-Ju Ran**, Quantum-Inspired Machine-Learning Methods Based on Tensor Networks, Tiangong University (17 Oct 2025).

[7] Invited talk (Chinese): **Shi-Ju Ran**, Quantum Many-Body Approaches Based on Variational Quantum Circuits and Tensor Networks, Chongqing-Chengdu Quantum Workshop (25 Aug 2025).

[8] Invited talk (Chinese): **Shi-Ju Ran**, Quantum-Inspired Machine Learning Based on Tensor Networks, CQCC 2025 Quantum Artificial Intelligence Forum, University of Electronic Science and Technology of China, Chengdu (21 Jul 2025).

[9] Conference paper: Sheng-Chen Bai, Kun Zhang, and Shi-Ju Ran, Quantum-Probabilistic Machine Learning: From Catastrophe of Orthogonality to Multi-qubit Encoding, CQCC 2025, CCIS 2733, 26-39, Springer (2026).

[10] Invited talk (Chinese): **Shi-Ju Ran**, Tensor-Network Machine-Learning Methods and Applications, Workshop and Summer School on Condensed Matter Physics, Computational Physics, and Emerging Interdisciplinary Topics, Henan Normal University, Xinxiang (23 Aug 2024).

[11] Invited talk (Chinese): **Shi-Ju Ran**, New Machine-Learning Numerical Methods for Strongly Correlated Quantum Systems, National Conference on Optoelectronic Science and Engineering, University of Electronic Science and Technology of China, Chengdu (27 Jul 2024).

[12] Invited talk (Chinese): **Shi-Ju Ran**, New Numerical Methods for Strongly Correlated Quantum Systems Based on Tensor Networks and Machine Learning, 18th National Conference on Magnetism Theory, Sichuan Normal University, Chengdu (9 Jun 2024).

[13] Invited talk (Chinese): **Shi-Ju Ran**, Machine Learning by Tensor Networks and Quantum Computing, "Invitation from Applied Physics, Exploring the Fascination of Theory" Strong Foundation Forum and Micro/Nanophotonics Academic Seminar XV,

Beihang University, Beijing (22 May 2024).

[14] Invited talk (Chinese): **Shi-Ju Ran**, Persistent Ballistic Entanglement Spreading with Optimal Control in Quantum Spin Chains, Frontier Interdisciplinary Workshop on Theoretical Physics, Nanjing Normal University, Nanjing (28 Apr 2024).

[15] Invited talk (English): **Shi-Ju Ran**, Persistent Ballistic Entanglement Spreading with Optimal Control in Quantum Spin Chains, 12th Workshop on Quantum Many-Body Computation (13 Apr 2024).

[16] Invited talk (Chinese): Shi-Ju Ran**, Quantum-Inspired Machine Learning by Tensor Networks, QuACT Frontier Lecture, Institute of Computing Technology, Chinese Academy of Sciences, Beijing (2023).**

[17] Invited talk (Chinese): **Shi-Ju Ran**, Tensor Networks for Efficient and Interpretable Machine Learning, Chinese Physical Society Fall Meeting, Ningxia University, Yinchuan (2023).

[18] Invited talk (Chinese): **Shi-Ju Ran**, Tensor Network Contraction, Summer School on Many-Body Computation, Minjiang University, Fuzhou (2023).

[19] Invited talk (Chinese): **Shi-Ju Ran**, Tensor Networks for Efficient and Interpretable Machine Learning, Workshop on Quantum Intelligence and Quantum Materials, Ludong University, Yantai (2023).

[20] Invited talk (English): **Shi-Ju Ran**, Tensor Networks for Efficient Quantum Computation and Many-Body Simulation, Chengdu-Chongqing Quantum Workshop - Quantum Sensing: Theory to Applications, Chengdu, China (2023).

[21] Invited talk (Chinese): **Shi-Ju Ran**, Efficient Simulation of Frustrated Quantum Spin Systems by Schmidt Tensor Network State Ansatz, 17th National Conference on Magnetism Theory, Hebei Normal University, Hebei (2023).

[22] Invited talk (English): **Shi-Ju Ran**, Efficient Quantum Computing by Optimal Control of Quantum Many-Body Dynamics, 11th Workshop on Quantum Many-Body Computation, Fuzhou, Fujian (2023).

[23] Invited talk (Chinese): **Shi-Ju Ran**, Functional Tensor Network Solving the Many-Body Schrodinger Equation, China Conference on Machine Learning and Applied Sciences, Beijing (2022).

[24] Invited talk (English): **Shi-Ju Ran**, Functional Tensor Network Solving the Many-Body Schrodinger Equation, 10th Workshop on Quantum Many-Body Computation, Jiangsu (2022).

[25] Invited talk (Chinese): **Shi-Ju Ran**, Classical or Quantum: Each to Its Own?, CCF Elite Conference on Quantum Computing, Beijing (2022).

[26] Invited talk (English): **Shi-Ju Ran**, Deep Learning Quantum States for Hamiltonian Prediction, Workshop II: Tensor Network States and Applications (online), IPAM, USA (2021).

[27] Conference paper: Wen-Jun Li, Zheng-Zhi Sun, Ya-Ru Wang, Shi-Ju Ran*, and Gang Su*, Matrix Product State for Quantum-Inspired Feature Extraction and Compressed Sensing, Second Workshop on Quantum Tensor Networks in Machine Learning, 35th Conference on Neural Information Processing Systems (2021).

[28] Invited talk (English): **Shi-Ju Ran**, Simulating Quantum Many-Body Systems by Few-Body Models at Zero and Finite Temperatures, Workshop on Tensor Networks in Many Body and Lattice Field, TDLI, Shanghai Jiao Tong University, Shanghai (2021).

[29] Invited talk (Chinese): **Shi-Ju Ran**, Deep Learning Quantum States for Hamiltonian Prediction, 16th National Conference on Magnetism, Yangzhou University (2021).

[30] Invited talk (Chinese): **Shi-Ju Ran**, New Machine-Learning Methods for Quantum Many-Body Physics, Summer School on Frontiers in Condensed Matter Theory, Hebei Normal University (2021).

[31] Invited talk (Chinese): **Shi-Ju Ran**, New Machine-Learning Methods for Quantum Many-Body Physics, YOCSEF Forum, Tianjin (2021).

[32] Invited talk (Chinese): **Shi-Ju Ran**, New Methods at the Intersection of Quantum Physics and Machine Learning, Third Academic Exchange Month, Beihang University (2021).

[33] Conference paper: Zheng-Zhi Sun, Shi-Ju Ran*, and Gang Su*, Tangent-Space Gradient Optimization: An Efficient Update Scheme for Tensor Network Machine Learning and Beyond, First Workshop on Quantum Tensor Networks in Machine Learning, 34th Conference on Neural Information Processing Systems (2020).

[34] Conference paper: Ding Liu, Zheng-Zhi Sun, Cheng Peng, Gang Su, and Shi-Ju Ran*, Generative Tensor Network Classification for Supervised Learning, International Workshop on Tensor Network Representations in Machine Learning, 29th International Joint Conference on Artificial Intelligence (2020).

[35] Invited talk (Chinese): **Shi-Ju Ran**, Bayesian Tensor Networks: Toward Efficient and Interpretable Probabilistic Machine Learning, First Workshop on Quantum Physics and Intelligent Computing (online), Tianjin-Beijing (2020).

Research Grants:

(1) National Natural Science Foundation of China (NSFC), Young Scientists Fund, Grant No. 12004266, "Efficient and Interpretable Quantum Machine-Learning Algorithms Based on Tensor Networks and Entanglement Theory," 1 Jan 2021 - 31 Dec 2023, completed, Principal Investigator.

(2) National Natural Science Foundation of China (NSFC), Key Program, Grant No. 11834014, "Development and Applications of New Computational Methods for Quantum Many-Body Systems," 1 Jan 2019 - 31 Dec 2023, completed, key participant.

(3) Beijing Natural Science Foundation, General Program, Grant No. 1232025, "Preparation and Tomography of Quantum Many-Body States Using Tensor-Network Machine-Learning Methods," Jan 2023 - Dec 2025, completed, Principal Investigator.

(4) Beijing Natural Science Foundation, General Program, Grant No. 1192005, "Simulation of Quantum Spin Liquids Using Tensor-Network Methods," Jan 2019 - Dec 2021, completed, Principal Investigator.

(5) Beijing Municipal Education Commission Science and Technology Program, General Project, Grant No. KM202010028013, "Quantum Machine-Learning Algorithms Based on Tensor Networks," 1 Jan 2020 - 31 Dec 2022, completed, Principal Investigator.

Student Supervision and Mentoring:

（1）Supervised five master's students to graduation;